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## **Statement of work in project**

The work contained in this project is that of the author and where material from other sources has been incorporated full acknowledgement is made.

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# 1 Methodology

This thesis investigates how machine learning and statistical modelling can be used to assess cost risk within a health insurance portfolio. The analysis is centred on two research questions. First, it examines how portfolio costs respond to stress scenarios such as increased hospitalisation rates and worsening chronic disease prevalence. Second, it evaluates how statistical and machine learning models can be used to estimate tail risk in annual medical costs.

The methodology proceeds in four linked phases. Phase 1 defines the portfolio and prepares the dataset for analysis. This includes cleaning the raw medical insurance data, identifying the annual medical cost variable as the modelling target, and constructing health-risk features related to hospitalisation, chronic disease burden and policyholder characteristics. Phase 2 conducts exploratory and diagnostic analysis to understand the distribution of medical costs. Particular attention is given to skewness, high-cost observations, tail percentiles and the relationship between health-risk factors and annual cost. This phase provides the justification for using both average-cost models and tail-aware modelling approaches.

Phase 3 develops and compares predictive models for individual-level annual medical costs. Ridge Regression is used as an interpretable statistical benchmark, while Gradient Boosting is used as a more flexible machine learning model capable of capturing non-linear relationships and interactions between risk factors. Several model specifications are considered, including grouped health-risk features, raw health features, interaction terms, smearing correction, robust loss functions and tail-weighted training. These models are evaluated using both general predictive accuracy metrics and tail-specific measures.

Phase 4 applies the fitted models to portfolio-level risk analysis. Predicted individual costs are aggregated to estimate total portfolio cost under baseline conditions and under stressed health-risk scenarios. Hospitalisation and chronic disease variables are altered to represent adverse changes in portfolio health, and the resulting changes in mean cost, total cost and upper-tail cost measures are examined. Quantile Gradient Boosting models are also used to estimate high conditional cost quantiles, including the 90th, 95th and 99th percentiles. This allows the methodology to assess not

only expected cost, but also the risk of unusually high claims.

The remainder of this methodology describes the data source, portfolio definition, feature engineering process, exploratory diagnostics, model specifications, evaluation metrics, tail-risk framework, stress-scenario design, model interpretation techniques and reproducibility controls used throughout the analysis.

## 1.1 Data Source and Portfolio Definition

The dataset used in this study was obtained from Kaggle and is titled *Medical Insurance Cost Prediction* [1]. The dataset contains policyholder-level medical insurance information for 100,000 individuals, including demographic, socioeconomic, lifestyle, health-status, insurance and claims-related variables. The response variable used throughout this dissertation is annual medical cost, which represents the yearly medical expenditure associated with each individual policyholder.

For the purposes of this study, the dataset is treated as a cross-sectional health insurance portfolio. Each observation represents one insured individual, and the full dataset represents the insurer's portfolio at a fixed point in time. The modelling objective is therefore not only to predict individual annual medical costs, but also to understand how changes in the health-risk composition of the portfolio affect aggregate cost outcomes.

Let  $Y_i$  denote the annual medical cost for policyholder  $i$ , where  $i = 1, \dots, n$ . The total observed portfolio cost is defined as

$$S = \sum_{i=1}^n Y_i, \quad (1)$$

where  $n$  is the number of policyholders in the portfolio. In the predictive modelling stage, the fitted models produce an estimated annual cost  $\hat{Y}_i$  for each policyholder. The predicted portfolio cost is then defined as

$$\hat{S} = \sum_{i=1}^n \hat{Y}_i. \quad (2)$$

This portfolio-level formulation is central to the first research question, which examines how the

insurance portfolio responds to adverse health scenarios such as increased hospitalisation rates and worsening chronic disease prevalence. By modifying selected health-risk variables and recalculating  $\hat{S}$ , the analysis measures the change in expected portfolio cost under stressed conditions.

The same framework is also used to address the second research question, which concerns the estimation of portfolio tail risk. Since health insurance costs are typically concentrated among a relatively small number of high-cost individuals, the analysis considers not only mean predicted cost, but also upper-tail cost measures such as the 90th, 95th and 99th percentiles. These quantities allow the models to be assessed in terms of their ability to identify and estimate unusually high medical costs, rather than only their average predictive accuracy.

The dataset therefore provides a suitable structure for investigating both individual-level cost prediction and portfolio-level risk behaviour. Individual policyholders form the unit of prediction, while aggregate and tail summaries of their predicted costs form the basis for stress-scenario and tail-risk analysis.

## 1.2 Data Cleaning and Feature Engineering

The raw dataset was imported into Python using a structured project workflow. The data were first loaded using `load_raw_data()` and then passed through `basic_clean()`. This initial cleaning stage standardised the column names into a consistent lower-case format, removed unnecessary spacing and ensured that variable names could be used reliably throughout the analysis. Duplicate records were also checked and removed where present. After cleaning, the dataset contained 100,000 policyholder observations and 58 variables after the engineered fields had been added.

The response variable for the analysis was `annual_medical_cost`. Since medical cost data are typically right-skewed, and the exploratory analysis confirmed that a small number of high-cost policyholders had a large influence on the distribution, a logarithmic transformation was created for modelling. For policyholder  $i$ , the transformed response was defined as

$$Y_i^{(\log)} = \log(1 + Y_i), \quad (3)$$

where  $Y_i$  denotes the original annual medical cost. The use of  $\log(1 + Y_i)$  reduces the influence of extreme costs during model fitting while preserving all observations, including low-cost policyholders. Predictions from models fitted on the transformed scale were later converted back to the original cost scale for interpretation and portfolio-level aggregation.

Feature engineering was then applied using `prepare_model_data()`. This added the transformed target variable `log1p_annual_medical_cost` and created grouped health-risk variables that were used throughout the exploratory analysis, model development and stress-scenario design. The main grouped variables were:

- `hospitalizations_grouped`, which grouped `hospitalizations_last_3yrs` into 0, 1 and 2+
- `chronic_count_grouped`, which grouped `chronic_count` into 0, 1, 2, 3 and 4+
- `days_hospitalized_grouped`, which grouped `days_hospitalized_last_3yrs` into 0, 1–3 and 4+

These grouped variables were created for two reasons. First, they reduce the influence of very sparse categories in the raw health variables. For example, only a small number of policyholders had two or more hospitalisations, and only a small number had four or more chronic conditions. Grouping these values creates more stable categories for modelling and interpretation. Second, the grouped variables directly support the first research question, because they provide interpretable health-risk states that can be modified under stress scenarios. For example, a hospitalisation stress scenario can be represented by moving policyholders from 0 to 1, or from 1 to 2+. Similarly, a chronic disease stress scenario can be represented by shifting policyholders into higher chronic disease burden groups. The model feature set was then constructed using `get_grouped_model_feature_columns()`. The primary grouped modelling specification contained 47 predictors before categorical encoding. These covered demographic characteristics, socioeconomic variables, lifestyle factors, clinical measurements, insurance plan characteristics, chronic disease indicators, procedure counts and the engineered health-risk groups. The identifier variable `person_id` was excluded from the modelling features because it does not contain predictive information about medical cost. Variables that directly describe

observed cost or claims outcomes, such as total claims paid or average claim amount, were also excluded from the primary feature set to reduce leakage from the response variable.

High-cost indicators were also engineered for the tail-risk analysis. After calculating empirical cost thresholds at the 90th, 95th and 99th percentiles, binary variables were created to indicate whether a policyholder exceeded each threshold. For quantile level  $q$ , the high-cost indicator was defined as

$$H_{i,q} = \begin{cases} 1, & \text{if } Y_i > Q_q, \\ 0, & \text{if } Y_i \leq Q_q, \end{cases} \quad (4)$$

where  $Q_q$  is the empirical  $q$ -th quantile of annual medical cost. These variables were not used as ordinary predictors in the main cost models. Instead, they were used to evaluate the extent to which different health-risk groups were associated with upper-tail costs and to assess model performance among high-cost policyholders.

Missing values were retained during the initial cleaning stage rather than removed, as deleting records would reduce the size and representativeness of the portfolio. Missing values were handled later within the modelling pipelines using imputation. This ensured that the same cleaned and engineered dataset could be used consistently across exploratory diagnostics, Ridge Regression, Gradient Boosting and tail-risk modelling.

The data cleaning and feature engineering process converted the raw Kaggle dataset into a structured health insurance portfolio suitable for both individual-level prediction and portfolio-level risk analysis. The engineered grouped variables provide the link between the raw health information and the stress-scenario research question, while the log-transformed target and high-cost indicators provide the foundation for modelling skewed costs and estimating tail risk.

### 1.3 Exploratory Diagnostics and Model Choice Justification

Exploratory diagnostics were conducted after data cleaning and feature engineering to assess the suitability of the dataset for predictive modelling, stress-scenario analysis and tail-risk estimation.

The purpose of this stage was to inform the methodological choices used later in the analysis.

The first part of the diagnostic analysis examined the response variable, annual medical cost. Sum-



mary statistics, histograms, density plots and box-plots were used to assess the shape of the cost distribution on both the original scale and the log-transformed scale. This was used to justify modelling  $\log(1 + Y_i)$ , where  $Y_i$  denotes annual medical cost for policyholder  $i$ . The transformed response was used in the main regression models because medical cost data are non-negative and commonly display asymmetry, with some policyholders having substantially higher costs than others.

The second part of the exploratory analysis examined the relationship between health-risk factors and annual medical cost. Grouped summaries and box-plots were produced for hospitalisation variables, chronic disease burden and days hospitalised. These diagnostics were used to assess whether the engineered health-risk groups provided interpretable categories for modelling and stress testing. In particular, the grouped variables `hospitalizations_grouped`, `chronic_count_grouped` and `days_hospitalized_grouped` were examined because they are directly related to stress scenarios. This ensured that the later stress tests were based on variables with a clear interpretation in terms of worsening portfolio health.

The third part of the diagnostic analysis focused on upper-tail medical costs. Empirical quantiles of annual medical cost were calculated and used to define high-cost policyholder indicators. These indicators allowed the analysis to examine whether different health-risk groups were associated with higher-cost regions of the portfolio. Tail concentration diagnostics were also used to assess the extent to which portfolio cost was concentrated among policyholders in the upper part of the cost distribution.

The exploratory diagnostics also informed the choice of modelling approaches. Ridge Regression was selected as the main statistical benchmark because it provides a regularised linear model that can handle a large number of encoded predictors while remaining relatively interpretable. This made it suitable for assessing the direction and relative strength of associations between policyholder characteristics and annual medical cost. However, a linear model may be limited when the relationship between health-risk factors and cost is non-linear or when interactions between variables are important.

Gradient Boosting was therefore selected as the main machine learning model. This method was chosen because it can capture non-linear relationships, threshold effects and interactions between

predictors without requiring these relationships to be specified manually in advance. This is particularly relevant for health insurance cost modelling, where the effect of one risk factor may depend on the presence of other risk factors. Gradient Boosting was therefore used to complement Ridge Regression by providing a more flexible predictive model for individual-level annual medical costs. The diagnostic focus on upper-tail costs also motivated the use of tail-aware modelling methods. Tail-weighted models were considered to give greater influence to high-cost observations during training. Quantile Gradient Boosting models were also fitted to estimate conditional upper quantiles of annual medical cost. These methods were included because average-cost models alone may not adequately describe the risk of unusually high claims, which is central to the estimation of portfolio tail risk.

#### 1.4 Statistical Modelling: Ridge Regression

Ridge Regression was used as the main statistical benchmark model for predicting individual annual medical costs. It was selected because it provides a regularised linear modelling framework that can handle a large number of predictors after categorical encoding while remaining more interpretable than a purely black-box machine learning model. In this context, Ridge Regression was used to establish a baseline for individual-level cost prediction and to provide a comparison point for the more flexible Gradient Boosting models.

The Ridge model was fitted to the log-transformed response variable. For policyholder  $i$ , the model can be written as

$$Y_i^{(\log)} = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad (5)$$

where  $Y_i^{(\log)} = \log(1 + Y_i)$ ,  $Y_i$  is the annual medical cost,  $\mathbf{x}_i$  is the vector of policyholder features,  $\beta_0$  is the intercept,  $\boldsymbol{\beta}$  is the vector of regression coefficients and  $\varepsilon_i$  is the error term. The log transformation was used to make the cost modelling problem more stable and to reduce the influence of extremely large observations during model fitting.

Unlike ordinary least squares, Ridge Regression estimates the coefficients by minimising a penalised loss function. The fitted coefficient vector is obtained by solving

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \left\{ \sum_{i \in \mathcal{T}} \left( Y_i^{(\log)} - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}, \quad (6)$$

where  $\mathcal{T}$  denotes the training set,  $p$  is the number of model predictors after preprocessing and  $\lambda$  is the regularisation parameter. The penalty term shrinks coefficient estimates towards zero, reducing over-fitting and improving stability when predictors are correlated or when many encoded categorical variables are included. The intercept term was not penalised.

A preprocessing pipeline was applied before model fitting. Numeric variables were imputed using the median and then standardised so that they had a comparable scale before the Ridge penalty was applied. Categorical variables were imputed using the most frequent category and then converted into dummy variables using one-hot encoding. Unknown categories in the test data were ignored during encoding to ensure that the fitted pipeline could be applied consistently to unseen observations. This preprocessing was embedded inside the model pipeline so that imputation, scaling and encoding were learned only from the training data and then applied to the test data.

The dataset was split into training and test sets using a fixed random seed to ensure reproducibility. The training set was used to fit the preprocessing pipeline, select the regularisation parameter and estimate the Ridge coefficients. The test set was held back for model evaluation. The regularisation parameter was selected using cross-validation over a predefined grid of candidate values. This allowed the strength of regularisation to be chosen from the data while avoiding direct tuning on the test set.

Several Ridge model specifications were considered to assess the sensitivity of the benchmark model to feature construction and tail-risk adjustments. The baseline Ridge model used the grouped health-risk features created during feature engineering. A wider regularisation grid was also considered to check whether model performance was sensitive to the range of candidate penalty values. A raw-health-feature specification replaced the grouped hospitalisation and chronic disease variables with their original forms, allowing the grouped and ungrouped representations to be compared. Another specification removed the grouped days hospitalised variable to assess whether this feature added useful information beyond the other health-risk variables. An interaction specification was

also created by combining hospitalisation group and chronic disease group into a single categorical interaction feature.

Because the model was fitted on the log-cost scale, predictions had to be transformed back to the original cost scale for interpretation. For a predicted value  $\hat{Y}_i^{(\log)}$ , the standard back-transformed prediction was calculated as

$$\hat{Y}_i = \exp\left(\hat{Y}_i^{(\log)}\right) - 1. \quad (7)$$

Predicted costs were clipped at zero where necessary so that the final cost predictions remained non-negative. A smearing correction specification was also considered to account for re-transformation bias when converting predictions from the log scale back to the original cost scale. The smearing factor was estimated from the training residuals on the transformed scale and then applied during back-transformation.

A tail-weighted Ridge specification was also fitted to investigate whether the statistical benchmark could be made more sensitive to high-cost observations. In this model, observations in the upper part of the training cost distribution were assigned larger sample weights during model fitting. The purpose of this adjustment was to give the model greater incentive to reduce prediction error for high-cost policyholders.

After fitting each Ridge specification, predictions were generated on the test set and converted back to the original annual cost scale. These predictions were then used in the model evaluation stage to assess average predictive accuracy, residual behaviour and tail performance. The Ridge models therefore provided an interpretable statistical baseline against which the Gradient Boosting and tail-aware models could be compared.

## 1.5 Machine Learning Modelling: Gradient Boosting

Gradient Boosting was used as the main machine learning approach for predicting individual annual medical costs. It was selected because it can model non-linear relationships and interaction effects between policyholder characteristics without requiring these relationships to be specified manually.

This is important in health insurance cost modelling because the effect of one health-risk factor may depend on other factors, such as age, chronic disease burden, hospitalisation history or lifestyle characteristics.

The Gradient Boosting model represents the prediction function as an additive ensemble of decision trees. For a policyholder with feature vector  $\mathbf{x}_i$ , the fitted model can be written as

$$\hat{f}(\mathbf{x}_i) = f_0(\mathbf{x}_i) + \sum_{m=1}^M \nu \gamma_m h_m(\mathbf{x}_i), \quad (8)$$

where  $f_0(\mathbf{x}_i)$  is the initial prediction,  $h_m(\mathbf{x}_i)$  is the decision tree fitted at boosting iteration  $m$ ,  $\gamma_m$  is the contribution of that tree,  $\nu$  is the learning rate and  $M$  is the total number of boosting iterations. Each tree is fitted sequentially so that later trees focus on the errors remaining from earlier trees. This allows the ensemble to build a flexible prediction function from a series of simpler models. As with Ridge Regression, the main Gradient Boosting models were fitted to the log-transformed response variable. For policyholder  $i$ , the response was

$$Y_i^{(\log)} = \log(1 + Y_i), \quad (9)$$

where  $Y_i$  denotes annual medical cost. Fitting the model on the transformed scale reduced the influence of extremely large cost observations during training. After prediction, fitted values were converted back to the original cost scale using

$$\hat{Y}_i = \exp\left(\hat{Y}_i^{(\log)}\right) - 1. \quad (10)$$

Predicted costs were clipped at zero where necessary so that all final predictions were non-negative. The same preprocessing pipeline that was applied before the Ridge Regression model was also applied here. Numeric variables were not standardised for the Gradient Boosting models, because tree-based models do not require predictors to be on a common scale. The preprocessing steps were fitted only on the training data and then applied to the test data, ensuring that information from the test set was not used during model training.

The dataset was split into training and test sets using the same fixed split used for Ridge Regression. This allowed the statistical and machine learning models to be compared on a consistent basis. The training set was used to fit the preprocessing pipeline and estimate the Gradient Boosting model. The test set was held back for model evaluation.

Several Gradient Boosting specifications were fitted to examine how model design affected average-cost prediction and tail-risk estimation. The baseline specification used the grouped health-risk variables created during feature engineering and was fitted using squared-error loss. A regularised specification was also considered, using a more constrained tree structure and subsampling to reduce over-fitting. A raw-health-feature specification replaced the grouped health-risk variables with their original forms, allowing grouped and ungrouped feature representations to be compared. A further specification removed the grouped days hospitalised variable to assess whether this information was useful beyond the other health-risk predictors. An interaction specification was also fitted by combining hospitalisation group and chronic disease group into a single categorical feature.

Alternative loss functions were used to assess whether the model could be made more robust to the distributional features of medical cost data. The squared-error loss was used for the main mean-cost prediction model. The absolute-error loss was considered because it is less sensitive to large errors than squared-error loss. A Huber loss specification was also fitted as a compromise between squared-error and absolute-error loss. This allowed the modelling strategy to compare standard mean prediction with more robust alternatives.

A tail-weighted Gradient Boosting model was also fitted to increase the influence of high-cost policyholders during training. In this specification, observations in the upper part of the training cost distribution were assigned larger sample weights. The purpose of this model was to assess whether giving more importance to expensive claims could improve prediction in the upper tail of the cost distribution.

Quantile Gradient Boosting models were fitted as a separate tail-risk modelling approach. Instead of estimating the conditional mean of annual medical cost, these models estimate conditional upper quantiles. For a quantile level  $\alpha$ , the model minimises the pinball loss

$$L_{\alpha}(Y_i, \hat{Y}_i) = \begin{cases} \alpha(Y_i - \hat{Y}_i), & \text{if } Y_i \geq \hat{Y}_i, \\ (1 - \alpha)(\hat{Y}_i - Y_i), & \text{if } Y_i < \hat{Y}_i. \end{cases} \quad (11)$$

This asymmetric loss penalises under-prediction and over-prediction differently, allowing the model to estimate a chosen upper point of the conditional cost distribution. The quantile models were therefore used to assess the risk of unusually high annual medical costs for individual policyholders and to support the portfolio tail-risk analysis.

After fitting each Gradient Boosting specification, predictions were generated for the test set and converted back to the original annual cost scale where required. The resulting predictions were used in the model evaluation stage to compare average predictive accuracy, residual behaviour and tail performance. The Gradient Boosting models therefore provided a flexible machine learning counterpart to Ridge Regression and formed the main modelling basis for assessing non-linear cost behaviour, high-cost policyholder risk and portfolio-level stress responses.

## 2 Introduction to Formatting and Academic Writing

### 2.1 Sections and Subsections

Welcome to your project, let's start with the basics of sectioning. Your project will be written using sections and subsections, and you would use the following commands `\begin{section}` and `\begin{subsection}`, if you want to cross-reference and refer to a section then use `\autoref` combined with a `label` at the corresponding section name. For example, Section 2 gives an overview of how to format an academic report.

### 2.2 Writing Equations

It may be the case that your project has some equations, all equations should be punctuated and be numbered. Let's consider an example and think about the circle, the area of circle is

$$A = \pi r^2, \quad (12)$$

where  $r$  is the radius of the circle and  $\pi$  is a constant and is equal to  $\approx 3.14$ . We can reference our circle equation in the text by using `\autoref{eq:circle}` and it appears as seen in Equation (12).

## 2.3 Inserting a Figure

You may have some Figures in your project. It is important that your Figures are big enough for the reader to see and that the axes labels are clear. You may therefore need to edit your Figures such that variable names do not contain underscores for example.

It is important that Figures and Tables have captions. The placement of captions is very important, for Figures, the caption goes underneath the Figure and for Tables the caption goes above. Stylistically you may wish to have your caption in italics (or one font size smaller if you prefer). In this document captions are in italic. We do this to ensure that the caption is separate from the main body of text. It is also clearer for the reader. If you are including any tables then the caption is above the Table.

Ideally Figures and Tables should be placed at the top or bottom of a page, they should not be sandwiched between text. Ideally, you should try to keep your Figure/Table contained within the appropriate section. Figures and Tables should be centred horizontally.



*Code 1: Code used to create Figure 1*

```
1 \begin{figure}[!tb]
2 \centering
3 \includegraphics[width=0.7\textwidth]{file name including the extension}
4 \caption{A picture of a beautiful sprocker spaniel! \label{fig:dog}}
5 \end{figure}
```

Let's now explain the code given in Code 1 `[!tb]` means that the Figure will be placed either at the top (that's what the `t` stands for) or bottom (that's what the `b` stands for) of the page. `\centering` means that the Figure will be in the centre of the page. Note that L<sup>A</sup>T<sub>E</sub>X may not always place your Figure where you want it but will place it in the first available place which satisfies your `[!tb]`.

`\label{fig:name}` is the general form label for the Figure and will be used to reference the Figure in the text. We would use `\label{tab:name}` for tables, `\label{eq:name}` for equations. Try to give an appropriate name for your labels. Note you cannot have more than one Figure/Table/Equation with the same label.

To reference the Figure in the text you would use `\autoref{fig:dog}` Using this, we can cross reference Figure 1 which shows a beautiful sprocker spaniel (cocker spaniel and English springer spaniel cross breed) in a sunny Paisley garden. You will notice that all links are clickable and will take you to the appropriate Figure.

## 2.4 Referencing

You will need to reference in this project and a lot of students under-reference. You will need to have a `.bib` file which stores all your references. If you speak to your supervisor they will show you how to make a bib file. You use the command `\cite{}` to cite references in your text and you separate your references with a common e.g.

`\cite{pritchard2010learning,harris2015mathematics}` would produce [2, 3] and these references would appear at the references section at the end of the document. As seen in Subsection 2.4



*Figure 1: A picture of a beautiful sprocker spaniel.*

## 2.5 Including Code

We can include code in the text using the commands displayed below

```
\lstinputlisting[language=R,  
caption=Insert caption here.  
\label{lst:Rcode1}]{codename.R}
```

The above results in code shown in Code 2. Remember we only include code in our document if we are going to refer to it in the text. We can either include a small R file for example, or type in the appropriate code.

*Code 2: Sample R code to create a bar plot.*

```
1 weight<-read.csv("WeightClassification.csv")
2 micro<-read.csv("gastro_microscope.csv")
3
4 x<-table(weight$BMI)
5 barplot(x, las=2)
```

## 2.6 Common mistakes in academic writing

The most common mistakes in academic writing are

- using casual or colloquial language;
- using contractions such as couldn't shouldn't, wouldn't or can't. These should all be replaced with could not, should not, would not and cannot;
- typos;
- having information in the wrong sections;
- writing in first person.

## 3 Referencing

You will need to reference in this project and a lot of students under-reference, particularly in the Methodology section. Generally, we use numerical references in the text and then the list the references in order of appearance in the referencing section. In other disciplines author surname and year will be given in the text and then the references will appear in alphabetical order in the reference list at the end of the document.

The document is set up for numerical referencing.

You need to remember to reference R or R packages you are using. If you have used the require command you will need to reference the package. You can use the citation command in R to get

your references easily e.g. `citation("palmerpenguins")` will produce the Harvard style reference for use in Word. We use the `\texttt{}` for R packages and commands.

To reference in L<sup>A</sup>T<sub>E</sub>X you will need to create a .bib file which contains all your references. You then use the `\cite{}` command to cite your work. For example, some interesting papers about pedagogy are [2] and [3]. Again, these references are fully clickable, and if you click on they will take you to the reference list at the end of the document.

## References

- [1] Mohan Krishna Thalla. Medical insurance cost prediction. Kaggle Dataset, 2025. Accessed: 2026-06-22.
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- [3] Diane Harris, Laura Black, Paul Hernandez-Martinez, Birgit Pepin, Julian Williams, and with the TransMaths Team. Mathematics and its value for engineering students: what are the implications for teaching? *International Journal of Mathematical Education in Science and Technology*, 46(3):321–336, 2015.